

Ministry of Education and Science of the Russian Federation

APPROVED BY

Director of IPN GPND

_____A. B. Pokhmelevsky

"__" _____2018

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GENERATOR OF SOFTWARE DOCUMENTATION AND RESEARCH REPORTS FROM THE
SOURCE FILES OF THE "MARKDOWN" FORMAT

Program description

APPROVAL

SHEETABVG.00099-01

13 01-LU

AGREED BY

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2019

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Description of the
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1 GAME DEVELOPMENT TECHNICAL TASK

- Genre: horror
- Setting: the underwater world - a flooded city after the explosion of the laboratory
- Main Feature: crafting

1.1 Description

This technical task describes the development of a horror game in an underwater setting, where the player explores a flooded city, collects resources and crafts improvements for survival. The world is inhabited by mutants, some of which live in closed oxygen bubbles, they are especially dangerous. There is an organization in the world engaged in the extraction of forgotten items and documents. The player interacts with its representatives and completes tasks. Crafting is a key mechanic: improving the harpoon, suit and balloons affects the gameplay and opens up new areas.

1.2 Tasks

- Create an underwater horror with a crafting system and progress of items.
- Implement the mechanics of limited oxygen.
- Work out several types of mutants and their behavior, including especially dangerous mutants living in oxygen bubbles.
- Create a crafting system. Collection of components, recipes. Upgrades of harpoon, suit, cylinders.
- Implement a mission system from an artifact and data mining organization.
- Implement inventory, survival interface (hud of oxygen, strength, pressure), preservation system.
- Develop a sound and visual atmosphere (voltage, radio interference, muffled sounds under water)

1.3 Functionality requirement

1.3.1 Game world

- Overall concept
 - Flooded city. Streets, underground passages, residential buildings, offices, industrial area, a large sunken laboratory.
 - Level of verticality. Above the city level, floating ruins and bubbles of acid at different depths.
- Filling:
 - Laboratory, archives, market, motor transport, underwater tunnels.
 - Resources in the form of metal, polymers, electronics, biomaterials (from mutants), rare drawings and files.
 - Hazardous currents, landslides, spikes, high-pressure zones, toxic emissions.
- Oxygen bubbles:
 - Occur in limited quantities. Inside, there are open areas without water, but mutants can live.
 - Bubbles serve for short rest and preservation.

1.3.2 Survival & Resources

- Oxygen:
 - Cylinders with limited volume.
 - Oxygen consumption depends on activity, suit condition and depth.
 - Replenishment of oxygen at the bases and stations of the organization through the crafting of new cylinders.
- Health:
 - Medicines can be crafted or found.
- Pressure and Temperature
 - At great depths, reinforced equipment is needed, otherwise gradual damage and breakage of the suit will occur.

1.3.3 Crafting & Progress

- Balanced Crafting System
 - Basic recipes are available immediately. Advanced in drawings and files from the laboratory.
 - Crafting components - metal, electronics, seals, biomaterials, catalysts.
 - Mobile kit for field repairs.
- Crafting Items:
 - Basic harpoon. Modifications: reinforced tips, explosive charges, electric charge, rope-electromagnet, the ability to use as an anchor.
 - Improvements in suit protection, pressure resistance, oxygen regeneration integration, module slots.
 - Cylinders: container, leakage rate, filters (from poisonous gases), air change module.
 - Auxiliary items such as flashlights, beacons, traps, markers, oxygen capsules, first aid kits.
- Pumping
 - Skill levels and points or progress through item upgrades and blueprints.
 - Each upgrade has the price of components and rare materials.

1.3.4 Enemies & Encounters

- Types of mutants:
 - Small: fast, attack in groups. Base materials
 - Medium: aggressive, use shelters, can cause panic effects.
 - Bubble: live in oxygen bubbles. Strong, often protected. Individuals go like bosses or mini-bosses.
 - Laboratory: mutants with hardware implants.
- Conduct:
 - Patrolling, responding to sound or light, hunting in packs.

- Battles
 - Importance of positioning in water: movement affects accuracy and speed.

1.3.5 Quests & Organization

- Organization:
 - Gives tasks: get documents, return samples, clean up locations.
 - Trader-engineer: purchase, repair, exchange of items and drawings.
- quests
 - Story missions: movement along the main line.
 - Side effects: searching for items, rescuing the crew, cleaning up.
 - One-time raids on particularly dangerous locations.
- Awards:
 - Resources, blueprints, upgrades, and access to new equipment.

1.3.6 Interface and Management

- Hud
 - Oxygen, health, suit strength, pressure.
 - Quick access slots for first aid kits, capsules, ropes and harpoon.
- Inventory:
 - Slot. Inventory limitation is mandatory.
 - Crafting windows and descriptions of items, recipes.
- Management:
 - Swimming: push, dive, float. anchoring.
 - Combat: throwing and firing with a harpoon, stabbing, chopping damage, using lochs.

1.3.7 And... the music.

- Underwater sound environment: muffled sounds, echo, pressure in the ears, squeaks of metal.
- Music: intense, ambient, scalable and responsive to the threat.

1.3.8 Saving and Progressing

- Savings
 - Auto seeding in safe bubbles or bases. Check points.
 - At high difficulty level, severe saving mode is optional (limited saves, permanent loss of items)
- Player progress:
 - Craft-tree, opening blueprints, equipment improvements, story progress.

RECORD OF REVISIONS

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